Bench wiring sheet - protocol emulator on the MachXO3L kit (3 Nov 2026)

Everything is **3.3 V logic**: the FPGA and ESP32 pins are not 5 V tolerant. One common ground between every board. Check each lead by continuity to the labelled pad before blaming the design (programs/read_pins.asm shows what the chip sees).

Column key. J4 hole = the numbered hole on the kit's header J4 where the lead goes. **Ball** = the FPGA's own name for that same connection (printed beside the hole, used in the constraints file): same wire, second name, for checking. **Engine pin** = the number the chip's programs use for that pin (0-19). **Signal** = what the wire carries (CS, SCK, SCL...); the far end is in the next column (ESP32 GPIO, UNO pin or 74HC595 pin).

1. Core: kit header J4 to the ESP32 host - fit these first, then run rerun.ps1 (command r). ID register should read 0x0022.

J4 hole	Ball	Chip signal	ESP32 GPIO	Notes
3	K12	CS	5	host SPI
5	M14	MOSI	23	host SPI
7	L14	MISO	19	host SPI
9	M15	SCK	18	host SPI
11	GND	GND	GND	common ground
4	K13	IRQ	4	high while any engine has data
6	N14	engine pin 0	16	engine A's UART out, ESP32 Serial2 RX
26	G15	engine pin 15	17	glitch filter test: the ESP32 sends pulses into the chip

J4 holes 1 and 2 are the kit's 3.3 V: fine for pull-ups, never joined to the ESP32's 3.3 V. Kit: **bash fpga/machxo3/build.sh load** (volatile SRAM only, never the NVCM). ESP32 on COM7: **esp32\upload_and_watch.ps1 -Sketch chip_test**.

2. I2C: engine pins 2 and 3 (rerun.ps1 -Cmd i with the ESP32 as slave, -Cmd n with the UNO R3)

J4 hole	Ball	Engine pin	Bus	ESP32 as slave (-Cmd i)	UNO R3 as slave (-Cmd n)
10	M16	2	SCL	GPIO 22	A5
13	L15	3	SDA	GPIO 21	A4

One 5.1 k pull-up from SCL to 3.3 V and one from SDA to 3.3 V (J4-1 or the UNO's 3.3 V pin, one source only). UNO ground to the common ground; the UNO sketch (arduino/i2c_slave_nano, COM11) switches its own 5 V pull-ups off. **Never both slaves at once** (both answer at 0x50): unplug A4/A5 for -Cmd i.

3. SPI master to the 74HC595: engine pins 4 to 7 (rerun.ps1 -Cmd s)

J4 hole	Ball	Engine pin	Signal	74HC595 pin
14	L16	4	SCK	11 SRCLK
15	K14	5	MOSI	14 SER
16	K16	6	MISO	9 QH' (serial out)
17	K15	7	CS / latch	12 RCLK

595 power: pin 16 VCC = 3.3 V, pin 8 GND, pin 13 OE = GND (outputs on), pin 10 SRCLR = 3.3 V (not clearing). LEDs on QA..QH = pins 15, 1, 2, 3, 4, 5, 6, 7, each through about 330 ohms to ground (long leg to the 595). Every transfer returns the previous byte through QH'; the run ends on 0x55, so QA (pin 15) = high. Before the rebuild the LEDs stayed dark with the chip side proven good: meter the four power pins first, then QA after a run, then the QH' lead.

4. The whole of J4 (for anything new)

J4 holes	Balls	Chip pins	Engine pins	Direction
6, 8, 10, 13, 14, 15, 16, 17	N14 N16 M16 L15 L16 K14 K16 K15	uio[0..7]	0-7	in and out (open drain able)
18, 19, 20	J14 H14 J15	uo_out[2..4]	8-10	out only
23, 24, 25	J16 H15 H16	uo_out[5..7]	11-13	out only
26, 27, 28, 29, 30	G15 G16 F15 F16 E15	ui_in[3..7]	15-19	in only
1, 2	3.3 V			
11, 12	GND			

Kit LEDs: P16 lights for IRQ, N15 is the heartbeat (they light when driven low). **Cyclone IV EP4CE6** (arrived 3 Nov): pin locations still to be chosen in fpga/cyclone4/chip_c4.qsf; the same eight leads of section 1 move across; set CHIP_HZ = 50e6 in the sketch (SPI goes to 4 MHz).

Order of work after replugging

1. Section 1 only. Load the kit, upload the sketch, rerun.ps1 (command r): ID 0x0022, 0 problems. 2. Add section 2: rerun.ps1 -Cmd i (ESP32 slave) or -Cmd n (UNO slave). 3. Add section 3: rerun.ps1 -Cmd s.